

CONVOLUTIONAL NEURAL NETWORKS For Autonomous Cars



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Part 1 of this article dealt with introduction to convolutional neural networks, training and simulation, image identification and depth estimation. This part covers the future of the networks including decision making and technology enablers.

The trainable, multi-layered structure of convolutional neural networks (CNNs) sets them apart from other neural networks. Their layers can be customised by adjusting their parameters to best fit their intended function. Neurons are constantly improving the accuracy of their outputs by learning from each piece of input data. This is particularly useful for applications in autonomous vehicles.

Capabilities of CNNs to distinguish both the presence and depth of obstacles makes them promising backbones for autonomous transportation. However, ethical decision making in the face of a collision still provides a considerable obstacle to the use of autonomous vehicles in everyday life as these are not yet proven to be safer than human drivers. On the other hand, these vehicles will promote sustainability by decreasing energy consumption and environmentally harmful emissions while reducing wear on vehicle parts. The more the research is done on CNNs in autonomous vehicles, the closer we will get to these vehicles as the main form of transportation.

Development of a sustainable driving environment

A mechanism for deciding the best route can be included in the neural network of each vehicle. This route optimisation and resulting decrease in traffic congestion is predicted to reduce fuel consumption by up to 4 per cent, thus reducing the amount of ozone and environmentally harmful emissions.

Each vehicle will have a set of com-

mands learnt for different driving situations. When efficiency and part preservation are priority, commands will be executed to minimise wear on the vehicle and reduce energy consumption by 25 per cent. For example, a human driver stuck in traffic might hit the accelerator and then the brake excessively to move every time the traffic inches forward. This causes excessive wear on the engine and brakes of the vehicle. However, an autonomous vehicle system would be optimised to either roll forward at a slow enough rate so that it does not collide with the vehicle ahead, or not move until there is enough free road available. This will decrease wear on the vehicle's brakes and engine while optimising fuel efficiency. As a result, the lifespan of each vehicle will extend, thus decreasing demand for new vehicles and vehicle parts. Fewer vehicles manufacturing means conservation of resources such as fuels burnt in factories and metals used in production.

Decision making: Needs to improve a lot!

Decision-making process for autonomous vehicles is complex and can sometimes fail to prevent crash in an unexpected or unpredictable situation. Autonomous vehicles still cannot consider numerous ethical factors including passenger safety, safety of the people outside the vehicle and human intervention. The network simply considers the driving command based on the scene features.

Many nuances of these ethical factors are pushed aside in favour of assurances that the human in the driver's seat will intervene and the car will not be required to take any action other than to alert the driver. However, in reality, it cannot be assumed that the drivers will have the time and focus to react, or that they will